

16-741 Mechanics of Manipulation – Project Proposal

Human-like Handwriting Using a 6-DOF Robot Manipulator

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Autopen systems, specialized machines to reproduce handwritten signatures in large volumes, have existed for decades. By constraining the end-effector to three degrees-of-freedom (DOF), the specialized hardware simplifies the handwriting replication problem.



Figure 1: An example of a presidential autopen system manufactured by Signascrip [1].

We propose enabling a general 6-DOF robotic manipulator to perform handwriting behaviors, mimicking a human's procedure more closely. The robot will be tasked with picking up a ballpoint pen and replicating a handwritten note on a given plane of defined position and orientation.

To do so, computer vision techniques will be used to convert images of arbitrary handwritten notes into a set of spline trajectories for the arm to replicate. A hierarchical control structure will be devised to smoothly transition between firmly gripping the writing implement, approaching the writing plane, and tracking the handwriting trajectories. Low-level positional and orientational tracking of the end effector will be accomplished using a feedback controller calculating forward and inverse kinematics with screw coordinates.

The resulting system will demonstrate the capabilities of a general manipulator to perform contact-rich tasks often simplified by specialized robotic hardware.

References

- [1] A. Picchi, "What is an autopen? Here's what to know about the devices used by presidents, writers and more.," *CBS News*, Jun. 05, 2025.
<https://www.cbsnews.com/news/what-is-an-autopen-president-biden-trump-signature/>